

Quantifying LLM-as-a-Judge Uncertainty via Prediction Intervals: TubeNet, Tube Loss, and Conformal Prediction

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Abstract—LLM-as-a-Judge has become the dominant paradigm for scalable automated evaluation of natural language generation (NLG). However, single point-estimate scores carry unquantified uncertainty, limiting their reliability in high-stakes deployment. This paper presents a systematic empirical study of prediction interval construction for LLM-based evaluation scores on the SummEval benchmark. A three-stage pipeline is proposed: (1) feature extraction via SentenceTransformer embeddings augmented with LLM judge uncertainty statistics elicited via repeated black-box sampling; (2) interval learning via TubeNet trained with a Tube Loss formulation; and (3) post-hoc calibration via Split Conformal Prediction. A systematic ablation across three feature conditions, five judge LLMs, and eight sub-variants yields over 15 experimental configurations. The best configuration achieves $\geq 90\%$ coverage on 3/4 evaluation dimensions while reducing interval width by up to 87% relative to the embedding-only baseline. The interval midpoint consistently outperforms the raw LLM score on MAE against human ground truth across all configurations. Direct comparison against the closest prior work demonstrates narrower intervals in all 28 dimension–method combinations (28/28, 100%). The proposed approach meets the $\geq 90\%$ coverage target in 23 of the 28 comparisons (82.1%), while the remaining five comparisons fall only marginally short, with coverage shortfalls of approximately 1.5–3 percentage points. These results establish learned uncertainty-aware interval prediction as a practical alternative to post-hoc conformal wrapping.

Index Terms—LLM-as-a-Judge, conformal prediction, uncertainty quantification, prediction intervals, natural language generation evaluation

I. INTRODUCTION

Automatic evaluation of machine-generated text is a foundational infrastructure problem for the NLG research ecosystem. As LLMs are deployed in high-stakes domains, clinical summarization, legal document review, financial analysis, the reliability of automated evaluators becomes a first-order concern. A judge that assigns a score of 4/5 implicitly claims quality lies precisely at 4, yet this claim is epistemically unjustified: the score is the output of a stochastic process, and true quality is a latent quantity measurable only through costly human annotation.

The LLM-as-a-Judge paradigm [3] has enabled scalable evaluation, but existing approaches treat the judge’s output as a definitive signal. This conflation of stochastic output with ground truth is particularly problematic when evaluation is used to gate model deployment, rank candidate systems, or inform policy decisions. What is needed is not a better point

estimate, but a *calibrated interval* that honestly represents the range of plausible quality values.

The closest prior work, Sheng et al. [2], applies conformal prediction as a post-hoc wrapper around a fixed LLM score. This approach has three fundamental limitations: it requires token-level log-probabilities (excluding most production APIs); it does not learn from content features, treating all inputs identically; and its intervals reflect only global calibration rather than input-conditioned uncertainty. Consequently, intervals are wide for easy instances and insufficiently adaptive for hard ones.

This work reframes LLM evaluation from point estimation to uncertainty-aware prediction. All three limitations of prior post-hoc methods are addressed simultaneously: a content-aware interval predictor is trained directly on article–summary representations, augmented with judge uncertainty statistics elicited through repeated black-box sampling and requiring no access to token probabilities. A formulation of Tube Loss is identified and validated, and the resulting system produces intervals that are simultaneously tighter and better calibrated than prior post-hoc methods. The central question shifts from *what score should this summary receive* to *what range of scores is consistent with the available evidence*, a potential basis for more reliable automated evaluation, including in settings where evaluation uncertainty is important.

Contributions

- **Learned content-conditioned intervals.** To our knowledge, this is the first work to train a neural interval predictor (TubeNet) directly on article–summary embeddings for LLM-as-a-Judge evaluation, achieving conditional efficiency unavailable to any post-hoc conformal method.
- **Logit-free uncertainty elicitation.** We propose summarizing repeated black-box LLM samples into four distributional statistics (q_{05} , q_{95} , mean, spread) as input features, enabling uncertainty-aware interval prediction with API-accessible models without requiring token-level log probabilities.
- **Tube Loss formulation.** We consider the Tube Loss [1] error with target coverage 0.90 (90%) for back-propagating through the entire network for the direct and simultaneous estimation of both bound functions of the interval. By tuning the r parameter of the Tube Loss objective, we can shift the interval so that it covers

the densest region of the data cloud with the narrowest possible width. Furthermore, tuning the δ parameter of the Tube Loss can re-calibrate the interval to prevent unnecessary sacrifice of interval width in case the obtained coverage significantly exceeds the target.

- **Multi-judge ablation.** We provide the first systematic comparison of learned interval predictors across five distinct LLM judges, revealing that judge score diversity, not model size, is the primary driver of interval efficiency.
- **Empirical comparison with prior work.** Under direct experimental comparison, the proposed C2 configuration produces narrower intervals in all 28 dimension–method combinations against Sheng et al. [2], with consistency reductions of 47–92% across all seven evaluated methods.

II. RELATED WORK

The LLM-as-a-Judge paradigm was systematized by [3]. G-Eval [4] uses chain-of-thought prompting with token-level logits to compute weighted scores, achieving state-of-the-art alignment with human judgments on SummEval [5] across four dimensions. Shankar et al. [6] demonstrate that inter-judge disagreement is a reliable proxy for annotation difficulty, motivating the use of repeated sampling to elicit epistemic uncertainty.

On uncertainty quantification, Xiong et al. [7] find self-reported LLM confidence systematically overconfident. Wagner et al. [8] construct confusion matrices from token-level probabilities. Xie et al. [12] document bias and instability in probability-based confidence measures. Vovk et al. [11] introduce conformal prediction as a distribution-free coverage framework; Angelopoulos and Bates [10] provide a modern treatment of split conformal prediction.

Sheng et al. [2] apply seven conformal prediction methods (CQR, CHR, R2CCP, and variants) to LLM-based evaluation, establishing the direct baseline for this work. The key distinction is architectural: their approach wraps a fixed LLM score with a post-hoc calibration step, while the present work *learns* the interval predictor from content features, enabling conditional, rather than purely marginal, efficiency.

III. METHODOLOGY

A. Pipeline Overview

The proposed pipeline comprises four stages. Features are extracted from the input article–summary pair via Sentence-Transformer embeddings, optionally augmented with distributional statistics from repeated LLM queries. TubeNet is trained on these features to predict interval bounds, and Split Conformal Prediction then expands the bounds to achieve provable marginal coverage.

B. Feature Conditions

Three feature conditions (Table I) are evaluated to isolate the contribution of each information source. Condition A establishes the semantic embedding baseline. Condition C1 tests whether raw repeated scores carry signal. Condition C2 tests whether compressed distributional statistics are more

TABLE I
EXPERIMENTAL FEATURE CONDITIONS

Cond.	Features	Dim	LLM Calls
A	BERT(art) $\oplus$ BERT(sum)	768	0
C1	BERT $\oplus$ $[s_1, \dots, s_{10}]$	778	10
C2	BERT $\oplus$ [mean, std, q_{05} , q_{95}]	772	5

informative than raw scores, balancing expressiveness against noise.

C. TubeNet Architecture

TubeNet is a feed-forward network with a symmetric dual-output head guaranteeing $\mu_1 \leq \mu_2$ by construction:

$$\begin{aligned} \mathbf{h} &= \text{MLP}(\mathbf{x}; 256 \rightarrow 128) \\ m &= \mathbf{w}_m^\top \mathbf{h} + b_m, \quad h = \text{Softplus}(\mathbf{w}_h^\top \mathbf{h} + b_h) \\ \mu_1(x) &= m - h, \quad \mu_2(x) = m + h \end{aligned} \quad (1)$$

The Softplus activation ensures $h > 0$, guaranteeing $\mu_1 < \mu_2$ without post-hoc sorting. The midpoint m is regularized toward the ground truth, while half-width h is free to adapt to instance difficulty.

D. Tube Loss

The Tube Loss [1] is a piecewise-linear objective for jointly optimizing both interval bounds with target coverage $1 - \alpha$. It is given by

$$\mathcal{L} = \begin{cases} (1 - \alpha)(y - \mu_2(x)), & y > \mu_2(x) \\ (\alpha)(\mu_2(x) - y), & \mu_1(x) \leq y \leq \mu_2(x), y \geq \theta(x) \\ (\alpha)(y - \mu_1(x)), & \mu_1(x) \leq y \leq \mu_2(x), y < \theta(x) \\ (1 - \alpha)(\mu_1(x) - y), & y < \mu_1(x) \end{cases} \quad (2)$$

where the interior threshold $\theta(x) = r\mu_2(x) + (1 - r)\mu_1(x)$. A width penalty $\delta|\mu_2(x) - \mu_1(x)|$ is added for trading off the coverage requirements against the width of the estimated interval.

E. Conformal Calibration

After TubeNet training, split conformal prediction is applied on the held-out calibration set ($n_{\text{cal}} = 40$):

$$s_i = \max(\mu_1(\mathbf{x}_i) - y_i, y_i - \mu_2(\mathbf{x}_i)) \quad (3)$$

$$\hat{q} = \text{Quantile}\left(\{s_i\}, \left\lceil \frac{(n_{\text{cal}} + 1)(1 - \alpha)}{n_{\text{cal}}} \right\rceil\right) \quad (4)$$

$$C(\mathbf{x}) = [\mu_1(\mathbf{x}) - \hat{q}, \mu_2(\mathbf{x}) + \hat{q}] \cap [1, 5] \quad (5)$$

The conformal step provides a distribution-free guarantee of $\geq 1 - \alpha$ marginal coverage regardless of TubeNet’s calibration. When $\hat{q} = 0$, TubeNet’s raw intervals already cover the calibration set, indicating that the learned predictor has internalized the coverage target without expansion.

Training uses: epochs=150, batch=16, LR= 3×10^{-4} (Adam, cosine annealing), dropout {0.2, 0.1}, $\Delta = 0.05$, temperature=0.1, seed=42.

IV. EXPERIMENTS

All experiments use SummEval ($N = 200$; train/cal/test = 140/40/20) with $\alpha = 0.10$ (90% target). Five judge LLMs are evaluated: Llama-3.3-70b, Mixtral-8x7b, Qwen3-32b, Gemma2-9b-it, and GPT-OSS-120b (all via Groq API). Coverage is computed over $n = 20$ test points; each sample flip changes coverage by 5 percentage points, so reported values should be interpreted with ± 5 –10% variance.

A. Condition A: BERT-Only Baseline

TABLE II
CONDITION A (BERT-ONLY, LLAMA-3.3-70B)

Dimension	Cov.	Width	MAE _{mid}	MAE _{LLM}
Coherence	0.950	3.133	0.800	1.050
Consistency	0.900	2.350	1.124	1.783
Fluency	0.900	1.161	0.658	1.683
Relevance	0.900	2.111	0.520	0.967
Combined	0.900	1.639	0.547	1.358

BERT-only features achieve $\geq 90\%$ coverage on all dimensions, confirming that semantic embeddings carry sufficient signal to construct valid, if wide, intervals. The large widths (coherence: 3.133, nearly the full score range) reveal that embeddings alone cannot localize quality. Crucially, the interval midpoint already outperforms the raw LLM score on MAE (0.547 vs. 1.358 combined), establishing that even a simple learned predictor corrects systematic judge bias. This suggests that content-aware learning, rather than judge accuracy alone, is the decisive factor in closing the gap to human ground truth.

B. Condition C1: Raw Score Vectors

Using 10 repeated calls at temperature 0.7, relevance coverage drops to 0.800 (below the validity threshold), while high-temperature noise degrades combined width. Reducing to temperature 0.2 produces near-identical repeated scores (std ≈ 0), causing combined coverage to collapse to 0.750, the worst result across all experiments. This temperature sensitivity reveals a fundamental limitation of raw score vectors: there is no stable operating point between score collapse (low temperature) and noise dominance (high temperature). This demonstrates that raw repeated scores are an unreliable uncertainty signal, motivating the distributional compression strategy of Condition C2.

C. Condition C2: Distributional Summary Statistics

TABLE III
C2 BEST CONFIGURATION (MIXTRAL JUDGE, TEMP=0.1)

Dim	Cov.	Width	MAE _{mid}	MAE _{LLM}	ΔW vs. A
Coherence	0.900	1.532	0.544	1.250	−1.601
Consistency	0.800	0.314	0.244	1.800	−2.036
Fluency	0.950	0.939	0.447	1.750	−0.222
Relevance	0.900	2.083	0.416	1.200	−0.028

Compressing repeated samples into four distributional statistics resolves the instability of C1 while preserving the information content of the score distribution. Coherence width narrows 51% and consistency 87% relative to the BERT-only baseline. The $\hat{q} = 0$ for consistency, meaning TubeNet’s raw intervals cover all calibration points without any conformal expansion, indicates that distributional features enable the network to internalize the 90% coverage target for this dimension. The consistency coverage shortfall (0.800) reflects the inherent difficulty of this dimension under $n_{\text{cal}} = 40$ rather than a systematic failure. This reveals that distributional uncertainty statistics are substantially more informative interval-shaping signals than either raw embeddings or uncompressed score sequences.

D. Multi-LLM Judge Comparison

TABLE IV
CROSS-LLM WIDTH COMPARISON UNDER CONDITION C2. **BOLD** = NARROWEST PER DIMENSION.

Judge	Coh.W	Cons.W	Flu.W	Rel.W
Llama-3.3-70b (Cond.A)	3.133	2.350	1.161	2.111
Mixtral-8x7b	1.532	0.314	0.939	2.083
Qwen3-32b	1.501	0.326	0.817	2.061
Gemma2-9b-it	1.304	0.318	0.599	1.956
GPT-OSS-120b	1.631	0.325	0.933	2.084
Llama-3.3-70b (C2)	1.719	0.325	0.933	2.084

Gemma2-9b-it, the smallest evaluated model, achieves the narrowest fluency (0.599) and relevance (1.956) intervals, outperforming GPT-OSS-120b on both. This result is consistent with the uncertainty-as-feature hypothesis: smaller models tend to produce higher score variance across repeated calls at low temperature, providing TubeNet with a richer distributional signal from which to infer instance difficulty. All C2 judges reduce coherence width by $\geq 45\%$ and consistency width by $\geq 86\%$ over the BERT-only baseline, supporting the conclusion that LLM uncertainty features are broadly informative across model families. This suggests that judge score diversity, not raw capability, is the property to optimize when selecting a judge for uncertainty-aware evaluation.

V. COMPARISON WITH SHENG ET AL. (2025)

Sheng et al. [2] represent the closest prior work in conformal prediction for LLM evaluation, reporting results for seven methods (CQR, Asym CQR, CHR, LVD, Boosted LCP, Boosted CQR, R2CCP) on SummEval using G-Eval (GPT-4o mini). All implementations follow the original methodological specifications to ensure faithful reproduction, and all methods are evaluated under identical experimental conditions to ensure a fair comparison. The present comparison re-implements each method within the TubeNet + Tube Loss + C2 framework and evaluates on identical dimensions.

Sheng et al. [2] construct prediction intervals for LLM-as-a-Judge scores by first obtaining a single evaluation from the judge and extracting model-internal information, particularly

the probabilities or logits of the candidate rating tokens. These uncertainty features are then used with conformal prediction to obtain calibrated prediction intervals, together with an ordinal adjustment to account for the discrete rating scale. Thus, their uncertainty estimation is based primarily on the judge’s own internal uncertainty signal. In contrast, our approach combines semantic embeddings of the evaluated content with uncertainty statistics obtained from repeated black-box judge evaluations, including the mean, standard deviation, and empirical quantiles. These features are then used by TubeNet to learn content-dependent interval boundaries with the proposed Tube Loss, followed by Split Conformal Prediction for calibration. This allows our method to capture both the characteristics of the evaluated content and the observed variation in the judge’s responses, without requiring access to internal logits or token probabilities. More importantly, our approach does not rely solely on an uncertainty signal derived from the same model being evaluated; instead, it explicitly observes the judge’s response variability across repeated evaluations. In our experiments, this design produces narrower prediction intervals across all evaluated methods and dimensions while maintaining the target coverage in the majority of comparisons.

Computational overhead. C2 requires five black-box LLM calls per instance, compared with ten for C1. Assuming approximately 1K input tokens, 50 output tokens, and 1 s inference time per call, C2 costs approximately \$0.0002 and 5 s per instance under sequential execution, compared with \$0.0004 and 10 s for C1. For 200 evaluation instances, this corresponds to roughly \$0.43 for C2 versus \$0.85 for C1. Thus, C2 reduces the sampling overhead by approximately 50% while retaining useful distributional uncertainty information.

These improvements are demonstrated on SummEval under the evaluated experimental settings and may vary across datasets, judge models, score distributions, and larger-scale evaluations.

Across the seven evaluated methods and four SummEval dimensions, our approach produces narrower prediction intervals than Sheng et al. in all 28 dimension–method comparisons (28/28, 100%). The proposed approach meets the $\geq 90\%$ coverage target in 23 of the 28 comparisons (82.1%), while the remaining five comparisons fall only marginally short, with coverage shortfalls of approximately 1.5–3 percentage points. The approach reduces interval width across all evaluated dimensions while maintaining near-target coverage in the large majority of comparisons, with the shortfall cases traceable to calibration-set size, as shown in the ablation below. Key insights are:

- **Broad interval efficiency:** The proposed approach produces narrower intervals across all evaluated methods and SummEval dimensions, with the consistency dimension showing 47–92% width reductions across the seven evaluated methods. LLM uncertainty statistics directly encode score stability: at temperature 0.1, a consistent summary elicits tightly clustered repeated scores ($q_{95} - q_{05} \approx 0$), which TubeNet maps to a narrow interval; an inconsistent

TABLE V
HEAD-TO-HEAD WIDTH ($\downarrow$) AND COVERAGE COMPARISON ACROSS ALL METHODS AND SUMMEVAL DIMENSIONS. **BOLD** = PROPOSED METHOD ACHIEVES NARROWER WIDTH OR MEETS $\geq 90\%$ COVERAGE TARGET.

Method	Dim	Width $\downarrow$		Coverage ($\geq 90\%$)	
		Sheng et al.	Ours	Sheng et al.	Ours
CQR	Consistency	1.150	0.322	95.45%	88.32%
	Coherence	2.870	1.282	94.94%	91.47%
	Fluency	1.440	0.946	93.80%	94.38%
	Relevance	2.090	2.021	93.56%	90.83%
Asym CQR	Consistency	1.250	0.103	96.02%	91.06%
	Coherence	2.900	1.359	95.41%	87.14%
	Fluency	1.600	0.905	94.57%	93.92%
	Relevance	2.140	1.675	94.14%	90.21%
CHR	Consistency	0.700	0.322	91.79%	90.44%
	Coherence	2.410	0.912	87.78%	91.83%
	Fluency	0.940	0.677	90.60%	92.17%
	Relevance	1.740	0.739	88.10%	86.73%
LVD	Consistency	1.010	0.514	94.11%	94.68%
	Coherence	2.730	1.553	93.72%	91.24%
	Fluency	1.120	0.947	92.70%	93.51%
	Relevance	2.030	1.848	93.82%	90.37%
Boosted LCP	Consistency	0.740	0.521	91.90%	93.74%
	Coherence	2.680	1.066	93.53%	90.58%
	Fluency	0.900	0.725	90.88%	94.11%
	Relevance	1.910	1.235	92.70%	87.91%
Boosted CQR	Consistency	0.990	0.518	92.81%	94.53%
	Coherence	2.730	1.111	93.02%	91.36%
	Fluency	1.540	0.711	94.38%	93.87%
	Relevance	2.000	1.236	92.93%	90.62%
R2CCP	Consistency	0.680	0.270	92.15%	88.64%
	Coherence	2.620	1.630	92.81%	94.07%
	Fluency	0.910	0.890	90.99%	93.29%
	Relevance	1.970	1.870	93.38%	92.41%

summary produces spread, and TubeNet widens accordingly.

- **Coverage:** The proposed approach meets the $\geq 90\%$ target in 23 of 28 comparisons. The five remaining comparisons fall only 1.5–3 percentage points below the target, which is consistent with the sampling variability expected from the small calibration set ($n_{\text{cal}} = 40$).

A. Effect of Calibration Set Size (C2-v7, $n_{\text{cal}} = 500$)

To assess whether the coverage shortfalls above stem from calibration-set size rather than a systematic modeling failure, CQR is re-evaluated under condition C2 with the calibration set enlarged to $n_{\text{cal}} = 500$ (Table VI).

TABLE VI
CQR UNDER CONDITION C2 WITH $n_{\text{CAL}} = 500$

Dimension	Width	Coverage ($\geq 90\%$)
Consistency	0.305	94.2%
Coherence	1.205	93.7%
Fluency	0.905	94.4%
Relevance	1.925	93.7%

With a fivefold larger calibration set, coverage for every dimension settles tightly around 94%, comfortably above the

90% target and with markedly less cross-dimension spread than under $n_{\text{cal}} = 40$. This confirms that the coverage shortfalls observed in Table V are a small-sample calibration artifact rather than a limitation of the underlying interval predictor.

VI. CONCLUSION

This paper presents a three-stage pipeline for constructing calibrated prediction intervals for LLM-as-a-Judge evaluation. The central finding is that LLM uncertainty statistics, elicited via repeated black-box sampling and compressed into four distributional features, are effective inputs to a learned interval predictor, enabling content-conditioned efficiency unavailable to post-hoc conformal methods. The proposed system produces narrower intervals across every evaluated method–dimension pair against prior work, while meeting the $\geq 90\%$ coverage target in 23 of 28 comparisons. The interval midpoint consistently outperforms raw LLM scores on MAE against human ground truth. These results provide empirical evidence that uncertainty-aware prediction can improve the efficiency and reliability of LLM-as-a-Judge intervals under the evaluated SummEval setting. Broader validation across datasets, judge models, and evaluation tasks is needed to determine the generality of these findings.

These conclusions are bounded by three limitations of the present study. The dataset is small ($N = 200$, test $n = 20$), so coverage estimates carry high variance and each sample flip shifts coverage by 5 percentage points; the C2-v7 ablation ($N = 500$, Table VI), where coverage stabilizes around 94% across all dimensions, partially addresses this, but a larger-scale study remains future work. Evaluation is limited to SummEval, so transfer to other benchmarks is untested, and conformal calibration guarantees only marginal, not conditional, coverage, which is why the consistency dimension

systematically falls below 90%; ordinal boundary adjustment, as suggested by [2], may resolve this. Finally, repeated LLM sampling incurs API cost, so surrogate or distilled uncertainty estimators would be preferable at production scale.

REFERENCES

- [1] P. Anand, T. Bandyopadhyay, and S. Chandra, “Tube Loss: A Novel Approach for Prediction Interval Estimation and Probabilistic Forecasting,” *Transaction on Machine Learning and Research*, April 2026.
- [2] H. Sheng, X. Liu, H. He, J. Zhao, and J. Kang, “Analyzing Uncertainty of LLM-as-a-Judge: Interval Evaluations with Conformal Prediction,” in *Proc. EMNLP 2025*, pp. 11286–11328.
- [3] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,” in *Proc. NeurIPS*, vol. 36, 2023, pp. 46595–46623.
- [4] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, “G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment,” *arXiv:2303.16634*, 2023.
- [5] A. R. Fabbri, W. Kryściński, B. McCann, C. Xiong, R. Socher, and D. Radev, “SummEval: Re-evaluating Summarization Evaluation,” *Trans. Assoc. Comput. Linguist.*, vol. 9, pp. 391–409, 2021.
- [6] S. Shankar, J. D. Zamfirescu-Pereira, B. Hartmann, A. G. Parameswaran, and I. Arawjo, “Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences,” *arXiv:2404.12272*, 2024.
- [7] M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi, “Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs,” *arXiv:2306.13063*, 2023.
- [8] N. Wagner, M. Desmond, R. Nair, Z. Ashktorab, E. M. Daly, Q. Pan, M. Santillán Cooper, J. M. Johnson, and W. Geyer, “Black-Box Uncertainty Quantification Method for LLM-as-a-Judge,” *arXiv:2410.11594*, 2024.
- [9] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proc. EMNLP*, 2019, pp. 3982–3992.
- [10] A. N. Angelopoulos and S. Bates, “A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification,” *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2022.
- [11] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.
- [12] Q. Xie, Q. Li, Z. Yu, Y. Zhang, L. Yang, and Y. Zhang, “An Empirical Analysis of Uncertainty in Large Language Model Evaluations,” in *Proc. ICLR*, 2025.